Project Title: “The Development of a Web-Based Delivery Management System with Fleet Dispatch, Real-Time In-Transit Tracking, and Digital Proof-of-Delivery for Gabriel Rebuyon Logistics ”

RESEARCHER

Maderal, John Ernest L.

Marimon, Glaiza Jade M.

Montillero, Francis Adrian Q.

Tumakay, Jannine Rose B.

Cabangal, Axl Shawn C.

October 2026

I. Title of the Study

Web-Based Delivery Management System with Fleet Dispatch, Real-Time In-Transit Tracking, and Digital Proof-of-Delivery

The title reflects the core functionalities of the system, which are designed to digitize the booking and dispatch of delivery trips, monitor cargo while it is in transit, and capture digital proof-of-delivery records that support client billing.

II. Introduction

Every trucking and logistics business, regardless of its size or reach, depends on three basic operating functions: accepting bookings for cargo, dispatching the right truck to haul it, and confirming that the cargo reached its destination so the delivery can be billed. Dispatch records, in-transit updates, and delivery paperwork serve as the factual basis for trip assignments, client billing, and driver accountability, and errors or delays in these records quickly translate into missed pickups, unpaid trips, and strained relations between the carrier and its clients (Madyanti et al., 2023; Saghaei, 2016). For trucking companies such as provincial haulers, where a small office staff coordinates a fleet of trucks spread across wide service areas, the way deliveries are booked, dispatched, and tracked directly shapes both the reliability of the service and the quality of everyday operations.

Despite the availability of affordable automation tools, many trucking companies continue to rely on phone calls, group chat messages, and paper delivery documents. Studies consistently describe this setup as error-prone and administratively heavy: bookings arrive through scattered channels, trip status is relayed manually and can be missed or forgotten, and physical proof of delivery is only processed after the driver returns, delaying billing and obscuring the true status of the fleet (Madyanti et al., 2023; Santoso, 2025). Web- and GPS-based delivery management systems have been shown to give dispatchers a centralized view of every shipment, allow drivers to transmit location and status updates in real time, and replace paper manifests with electronic proof-of-delivery records that can be verified automatically (Kurniawan et al., 2025; Hiya et al., 2025; Renushe et al., 2026).

Gabriel Rebuyon Logistics currently operates a fleet of twelve trucking units composed of one six-wheeler straight truck with a drop side, five ten-wheeler straight trucks, and six prime movers, manned by approximately thirty drivers and staff members. The company hauls stock items for Gaisano, container vans for Trans-roadway Forwarders, water for Nature's Spring, and cargo for Pisac and cash customers, with delivery routes covering Butuan, Cabadbaran, Valencia, Malaybalay, Iligan, Tubod, Davao, Bukidnon, El Salvador, Lugait, Marawi, Pagadian, Ozamiz, and Zamboanga. Bookings, however, are received through different channels per client, tracking is handled manually through group chat updates or direct phone calls to drivers, and drivers submit physical delivery paperwork only upon returning to the office before billing can be processed.

This study proposes the development of a Web-Based Delivery Management System (W.D.M.S.), a cross-platform application designed to digitize booking documentation, automate trip dispatch, monitor deliveries in transit, and capture digital proof-of-delivery for the company. The system integrates a Fleet Dispatch and Tracking Module that assigns cargo to available units and records status updates from dispatch to delivery, real-time in-transit tracking based on driver smartphones and available dashcam GPS, and role-based interfaces that allow the administrator and dispatcher to manage bookings and validate delivery records while drivers update trip status and upload proof of delivery. The system is developed under the System Development Life Cycle using the Modified Waterfall Model, following a three-tier architecture composed of a presentation layer, an application layer, and a data layer.

Through W.D.M.S., the study aims to significantly reduce the time spent coordinating and monitoring deliveries, minimize lost paperwork and unrecorded trips, and provide transparent, organized, and accessible delivery records for both the management and the drivers of Gabriel Rebuyon Logistics. The remainder of this paper presents the statement of the problem, the objectives of the study, its significance, the scope and delimitations, the methodology, the review of related literature, and the expected outputs of the proposed system.

III. Statement of the Problem

The existing delivery management process of Gabriel Rebuyon Logistics is carried out almost entirely by hand. Bookings arrive as physical bills of lading, emailed documents, or online release orders depending on the client, after which the dispatcher assigns trucks and monitors trips through group chat messages and phone calls, and drivers submit physical delivery paperwork only upon returning to the office so billing can be processed. There is no dedicated system that links booking records to dispatch assignment and delivery confirmation, and there is no centralized platform where the status of every shipment and unit can be seen. The problems of this study therefore center on three specific parts of this existing process:

1. Booking and Documentation. The booking stage depends on varied client channels, with Nature's Spring issuing physical bills of lading, Pisac issuing online release orders, and Trans-roadway sending bills of lading by email. Because these documents are recorded manually, trip details can be missed, misplaced, or duplicated, and every booking must be re-encoded before a truck can even be assigned.
2. Dispatch and In-Transit Tracking. The company does not use centralized GPS for its trucks, and tracking is handled manually through group chat updates or direct phone calls to drivers. When a prime mover breaks down, a replacement truck head must be dispatched to take over the trailer, and locating the affected unit depends entirely on whoever answers the phone, delaying response times and leaving management without a live view of the fleet.
3. Proof of Delivery and Billing Records. Delivery confirmation relies on physical paperwork that drivers submit only after returning from a trip, so billing cannot begin until documents are handed in and processed. The existing setup lacks a streamlined, accessible platform where delivery records and proof of delivery are captured digitally at the point of delivery, and disputes over delivered or undelivered cargo are difficult to settle because no time-stamped digital record is kept.

Thus, there is a need for an automated system that digitizes booking documentation, automates fleet dispatch, provides real-time in-transit tracking, and captures digital proof-of-delivery records that support timely and accurate billing.

IV. Objectives of the Study

To develop and implement a Web-Based Delivery Management System (W.D.M.S.) that automates fleet dispatch, tracks deliveries in real time, and digitizes proof-of-delivery for Gabriel Rebuyon Logistics.

Specific Objectives:

1. To gather essential information regarding the existing manual delivery and dispatch workflow through targeted interviews with management, dispatchers, and drivers.

2. To design a multi-tier, cross-platform system utilizing a Presentation layer, Application layer, and Data layer.

3. To develop user interfaces providing specific functionalities for different roles, enabling drivers to update trip status and upload digital proof-of-delivery, while allowing the administrator and dispatcher to manage bookings, assign units, and monitor the fleet in real time.

4. To implement a backend Fleet Dispatch and Tracking Module that automatically records trip assignments, in-transit status updates, and digital proof-of-delivery, and organizes them into delivery records ready for billing.

V. Significance of the Study

This project will benefit the following stakeholders:

Drivers and Staff – will gain access to a simple interface where they can receive trip assignments, report delivery status, and upload proof of delivery without juggling separate group chats and paperwork.

Management and Dispatcher – will save significant time and effort by managing bookings in one place, assigning trucks to available units, monitoring deliveries in transit, and validating delivery records for billing.

Gabriel Rebuyon Logistics – will benefit from reduced lost paperwork, improved fleet visibility, faster billing cycles, and highly organized delivery records across its twelve units and client accounts.

Future Researchers – may use this study as a reference for developing similar delivery management and fleet tracking systems for other small and medium-sized trucking companies.

VI. Scope and Delimitation

This project focuses on the development of a cross-platform Web-Based Delivery Management System utilizing both a frontend application and a backend server for Gabriel Rebuyon Logistics. The scope encompasses the comprehensive management of client bookings, truck and driver records, delivery assignments, in-transit status updates, and digital proof-of-delivery, all of which are accessible through two primary user roles: Administrator/Dispatcher and Driver. The system's core capabilities include booking documentation for varied client channels, fleet dispatch assignment, real-time in-transit tracking through driver smartphones and available dashcam GPS, and the capture and validation of digital proof-of-delivery that feeds the billing process. Through these capabilities, the system is expected to replace the company's group chat and paper-based workflow with a centralized database in which every booking flows directly into dispatch and every delivery generates its own verifiable digital record, giving management a single point of control over the fleet while giving each client's cargo a documented chain of custody from the container yard to the destination.

To establish clear project boundaries, the system's logic and design are delimited specifically to the operational workflow of the company's twelve trucking units, its approximately thirty drivers and staff members, and the delivery routes covered by the study. The in-transit tracking feature relies on the GPS capability of the drivers' smartphones and the company's existing dashcams rather than dedicated GPS hardware installed on every unit, and locations are updated only while a device has connectivity. The system likewise does not cover functions beyond delivery management, such as route optimization, vehicle maintenance scheduling, warehouse and inventory management, and human resource management, and it does not automatically compute or remit driver compensation; per-trip records remain as references for the company's existing per-trip payment process. Finally, digital proof-of-delivery submissions are validated and approved by the administrator rather than automatically accepted by the system, ensuring that management retains final judgment over any delivery record that feeds client billing.

VII. Methodology

The study will adopt the System Development Life Cycle (SDLC) utilizing the Modified Waterfall Model as the primary development approach, ensuring clear requirement definitions while allowing minimal revisions between stages.

Development Tools:

- Frontend: Flutter
- Backend: Node.js with Express
- Database: MySQL / PostgreSQL

Research Process:

1. Requirements Gathering and Analysis – Conducting interviews with management, dispatchers, and drivers to identify current booking channels, dispatch practices, tracking methods, and challenges within the manual delivery process.

2. Design – Translating requirements into system models, including a Context Diagram, Use Case Diagram, Entity-Relationship Diagram (ERD), and a Data Flow Diagram (DFD), structured under a three-tier architecture.

3. Development – Implementing the system using the selected technology stack (Flutter, Node.js, MySQL and PostgreSQL) and developing the central Fleet Dispatch and Tracking Module.

4. Testing and Evaluation – Validating the cross-platform capabilities, RESTful API communications, and testing the logic of automated dispatch assignment, real-time status updating, and digital proof-of-delivery generation.

VIII. Review of Related Literature

This chapter presents reviewed studies, both foreign and local, that bear on the development of a web-based delivery management system with fleet dispatch, real-time tracking, and digital proof-of-delivery. The reviewed works are grouped into foreign literature, which covers studies conducted outside the Philippines, and local literature, which covers studies conducted within the country.

Foreign Literature

Madyanti et al. (2023) designed a web-based goods delivery information system to address companies that manually record and store shipment data, a practice that causes errors and slows down processing. Developed through the Waterfall model, the system manages shipment activity data automatically and in real time, generates shipping documents such as waybills and invoices automatically, and displays real-time shipment reports through a dashboard on the application's main page. The study demonstrates that even a straightforward web application can replace error-prone manual shipment recording while accelerating document processing, supporting the present study's decision to centralize bookings, dispatch records, and delivery documents in a single web-based platform.

Santoso (2025) developed a web-based goods delivery monitoring application using PHP, MySQL, and Bootstrap to address the limited visibility and real-time monitoring in the goods delivery process. The application provides real-time tracking features, shipment activity log recording, and a responsive interface accessible across various devices. The study's results indicate that the application improved operational efficiency, reduced errors in the delivery process, and provided better transparency for users, illustrating how a lightweight, browser-based monitoring layer can give dispatchers live visibility over deliveries without requiring complex enterprise deployments.

Kurniawan et al. (2025) developed a real-time delivery tracking application for furniture logistics that integrates WebSocket-based bidirectional communication, location-based services, and the Haversine formula for geodesic distance validation. The system comprises an Android driver application, a Laravel-based backend with real-time WebSocket infrastructure, and monitoring interfaces for administrators and customers, and its proof-of-delivery triggering is restricted to authorized geofence boundaries that achieved 100% validation accuracy across three test scenarios. Average WebSocket latency of 44.03 milliseconds, 23 to 68 times faster than traditional HTTP polling, supports the present study's premise that real-time driver-to-office status transmission is technically feasible even for small logistics operations.

Hiya et al. (2025) designed a real-time logistics fleet tracking system integrating Internet of Things technology with a web-based platform to address the lack of accurate fleet visibility in small and medium logistics operations. Built with an ESP32 microcontroller and GPS module

transmitting location data to a server visualized through a responsive web application, the system was developed using the Agile-Scrum methodology and evaluated through functional testing, a System Usability Scale score of 78.3, and user acceptance testing that confirmed alignment with operational requirements. The study shows that continuous vehicle location reporting materially improves tracking accuracy, operational efficiency, and service transparency, directly informing the tracking features of the present system.

Renushe et al. (2026) proposed a smart QR code-enabled transporter authentication and proof-of-delivery system integrated with real-time GPS tracking to address unauthorized transporter access and disputes over proof of delivery in the logistics sector. Unique, time-bound QR codes authenticate transporter personnel at loading and delivery points while GPS tracking provides live location updates throughout transit, and experimental evaluation with 100 users recorded a 92% reduction in fraud, 98.5% authentication accuracy, and a 97% delivery success rate. The study confirms that digitizing identity verification and delivery confirmation strengthens accountability and transparency, which the present system adopts in its digital proof-of-delivery workflow.

Local Literature

Lopez et al. (2025) developed an IoT-based fleet monitoring system with driver logging and OBD-II diagnostics for a Philippine tourist transport services company, integrating GPS, OBD-II telemetry, and cloud-based dashboards to enhance visibility, maintenance, and operational decision-making for small and medium-sized transport enterprises. Evaluated through the V-Model SDLC across ten vehicles, the system increased fleet utilization from 76.0% to 85.3%, decreased downtime from 24.0% to 14.7%, and improved fuel efficiency by 5.3%, while employing encryption and authentication measures compliant with the Philippine Data Privacy Act. The study demonstrates that locally developed fleet telemetry systems deliver measurable operational gains for small and medium transport operators, supporting the present study's goal of real-time fleet visibility.

Cueto (2026) examined the outbound warehouse dispatch process of a tissue manufacturing company in the Philippines and found that manual loading activities and the lack of real-time verification were the main causes of dispatch mistakes leading to significant

financial losses. Using a simulation-based Lean Six Sigma approach, the researcher designed a digital dispatch verification system that uses barcode scanning and structured loading controls, reducing dispatch discrepancies by 82.35% and related costs by 83.48% in simulation. The study provides local evidence that digital verification at the dispatch point prevents costly delivery errors, a principle the present system extends to truck dispatch and delivery confirmation.

Nuevas et al. (2024) developed a mobile and web application for Android that utilizes GPS services to improve incident emergency response in Abuyog, Leyte, enabling users to quickly report incidents, pinpoint their location, and notify the relevant authorities. The system was developed through an iterative process incorporating client feedback and was evaluated by thirty end-users, yielding a high acceptance rate reflected in an overall grand mean score of 4.77 for system quality and performance. The study shows that GPS-based mobile reporting can function reliably in Philippine provincial settings, informing the present system's use of smartphone-based location reporting for field operations.

Cabugatan et al. (2025) developed the GEOGAS App, a web application with geolocation technology that enhances the delivery of liquefied petroleum gas in the Municipality of Burauen, Leyte, using a route optimization algorithm based on customers' pinned locations. Built through the Agile SCRUM methodology, the system provided optimized and detailed geolocation maps of customer orders, and its software quality evaluation using ISO 25010 attained a rating of 3.95, interpreted as very effective, with unified positive feedback from delivery staff, office staff, and customers. The study illustrates how geolocation-based delivery applications can serve local delivery businesses in the Philippines, reinforcing the feasibility of the present system's tracking and dispatch features.

Cordova et al. (2025) examined how digital innovation contributes to entrepreneurial success in the Philippine MSME sector and identified the leading obstacles to digital adoption: cultural resistance, lack of technical expertise, and financial constraints. Their mixed-method study found that while MSMEs readily embrace social media marketing and digital payments, they have not leveraged more advanced technologies such as cloud computing and enterprise systems, and the authors recommend affordable tools, training, and awareness-building. This finding mirrors the situation of Gabriel Rebuyon Logistics, whose drivers are not tech-savvy and

whose management prefers its current manual billing process, underscoring the need for the present system to prioritize simplicity, training, and gradual adoption.

Taken together, the foreign and local literature consistently shows that centralizing delivery bookings, dispatch, and tracking in a web-based system improves fleet visibility, reduces documentation errors, and strengthens accountability through digital proof of delivery. However, the reviewed systems were developed for other organizations with different fleet sizes, client arrangements, and operating regions, and none covers the operations of Gabriel Rebuyon Logistics. This gap motivates the development of W.D.M.S., which is tailored to the company's twelve trucking units, its client-specific booking channels, its Mindanao-wide delivery routes, and its existing per-trip compensation and billing practices.

IX. Expected Output

The final output will be a fully functional Web-Based Delivery Management System (W.D.M.S.) with the following deliverables:

- A cross-platform frontend application serving dispatchers, administrators, and drivers smoothly across supported devices.
- A centralized backend system managing fleet dispatch and delivery records, successfully integrating bookings, trip assignments, in-transit status updates, and proof of delivery.
- Automated generation of delivery summaries and digital proof-of-delivery records that support timely client billing.
- A delivery status workflow in which drivers submit in-transit updates and proof of delivery, and the administrator validates them.

The system is expected to greatly reduce the time spent coordinating and monitoring deliveries and eliminate lost paperwork and unrecorded trips.

X. References

Cabugatan, R. J., Esponga, C., Giguracion, M. G., Refuerzo, M. A., de los Santos, J. R. N., de los Santos, G. N., & De Los Santos, L. G. (2025). GEOGAS App: A geo routing LPG delivery service application. https://doi.org/10.1109/it64745.2025.10930303

Cordova, W., Lim, C. T. N., Ong, H. T., Caraan, H. S., Dela Cruz, G. A. R., & Jimenez, E. (2025). Digital innovation and entrepreneurial success in the Philippine MSME sector: Challenges and opportunities. Economics. https://doi.org/10.2478/eoik-2025-0056

Cueto, N. D. (2026). Simulation-based optimization of the outbound warehouse dispatch process to minimize risk of delivery discrepancies. https://doi.org/10.46254/ap07.20260130

Hiya, G. E., Tony, T., & Sitorus, M. D. L. (2025). Design of a web and Internet of Things-based logistics fleet tracking application. Internet of Things and Artificial Intelligence Journal, 5(4). https://doi.org/10.31763/iota.v5i4.1057

Kurniawan, B., Mumpuni, R., & Al Haromainy, M. M. (2025). Design and development of a real-time delivery tracking app using WebSocket and Haversine. bit-Tech, 8(2). https://doi.org/10.32877/bt.v8i2.3275

Lopez, S. K. R., Tipon, J. J. D., Visbal, J., Castro, A. B., & Guarin, B. (2025). IoT-based fleet monitoring system with driver logging and OBD-II diagnostics. https://doi.org/10.1145/3787330.3787339

Madyanti, A. N., Ananda, N., Rini, M. W., & Rizal, S. (2023). Design of a web based information system to manage delivery activities. Jurnal Rekayasa Sistem Industri, 10(2). https://doi.org/10.25124/jrsi.v10i02.721

Nuevas, L. K., Velarde, J. L. C., Javellana, J. B. A., & Torlao, J. R. E. (2024). Real-time incident reporting and emergency response: A mobile GPS app for Abuyog, Leyte, Philippines. https://doi.org/10.1109/icvee63912.2024.10823942

Renushe, D. H., Dhang, A. R., Deshmukh, D., Patil, D., & Rasal, P. R. (2026). Smart QR code-enabled transporter authentication and proof of delivery system with real-time GPS tracking. The Bioscan, 21(2), 494–501. https://doi.org/10.63001/tbs.2026.v21.i02.pp494-501

Saghaei, H. (2016). Design and implementation of a fleet management system using novel GPS/GLONASS tracker and web-based software. arXiv preprint arXiv:1610.02667. https://doi.org/10.48550/arxiv.1610.02667

Santoso, M. F. (2025). MF optimization of delivery management through a web-based monitoring application using PHP and Bootstrap. Jurnal Teknologi dan Sistem Informasi Bisnis, 7(3). https://doi.org/10.47233/jteksis.v7i3.1686

XI Appendices

Appendix A – Consolidated Data

Sample Questionnaire:

Business Profile & Fleet Baseline

- How many active trucks do you currently operate, and what types are they (e.g., 4-wheeler, 6-wheeler, 10-wheeler, wing van, flatbed)?
- Who are your primary clients (e.g., walk-in individuals, regular corporate contracts, agricultural/wholesale distributors)?
- How many total staff handle the daily logistics (number of dispatchers, full-time drivers, freelance/on-call drivers)?
- What is the typical geographic coverage of your deliveries (e.g., local city distribution, provincial, or inter-island via RoRo)?

Current Booking & Transaction Workflow

- Step-by-step, how does a customer request and confirm a booking right now (e.g., phone call, text message, Facebook page, walk-in)?
- How do you record and track active bookings (e.g., physical logbook, whiteboard, Excel sheet)?
- How are pricing and rates determined (e.g., fixed rate per truck type, flat distance fee, weight-based, or negotiated case-by-case)?
- What is the current payment process (e.g., 50% downpayment upon pickup, full cash upon delivery, post-delivery invoicing)?

Dispatching & In-Transit Visibility

- How do you currently assign a driver and a truck to a specific booking? What criteria decide who gets the trip?
- Once the truck leaves, how do you and the customer know its location and arrival time (e.g., calling the driver while driving, SMS updates)?
- How do drivers submit Proof of Delivery (POD) (e.g., paper waybill signed and returned days later, photos sent on messaging apps)?
- What happens when an issue occurs on the road (e.g., breakdowns, heavy traffic, route diversions, bad weather)? How is that communicated?

Operational Pain Points & Delays

- What is the single biggest cause of operational delays or lost revenue in your current setup?
- Have you experienced double-bookings, misplaced waybills, or unassigned idle trucks? How often does this happen in a month?

- How much time does the dispatcher or owner spend each day just answering calls asking, "Where is the truck right now?"

Interview Transcripts:

Excerpt 1

"Ang atoang units kay straight truck 10-wheeler, straight 6-wheeler na dropside, tapos naay prime mover heads na trailer remover para sa 20-footer ug 40-footer container vans. Ang clients nato Gaisano, Nature's Spring, ug mga trader. Sa Gaisano, gikan sa Manila ang container van pagbaba sa barko, i-send nila sa email usahay 80 ka BLs (Bills of Lading), unya kami magkarga gikan sa port o bodega."

Excerpt 2

"Ang problema gyud usahay kung mangutana ang client kung asa na ang truck. Naa man GPS ug dashcam pero walay portal ang client, so sige silag tawag. Samot na kung magubaon sa dalan—kung prime mover, ulo ra ang i-rescue; pero kung straight truck, kailangan og laing truck para ibalhin ang karga. Kinahanglan gyud makita dayon asa dapit na-aberya."

Excerpt 3

"Ang bayad sa driver kay 10% per trip example kung ₱80,000 ang trip rate, ₱8,000 ang share sa driver. Ang problema sa proof of delivery kay papel. Ibilin pa sa bodega, unya madala ra balik diri human sa pila ka adlaw. Kung naay cellphone app na piktyuran lang dayon ang pinirmahang papel, mas paspas ang billing sa Net 7 o Net 30 terms."

Appendix B – Use Case Diagram

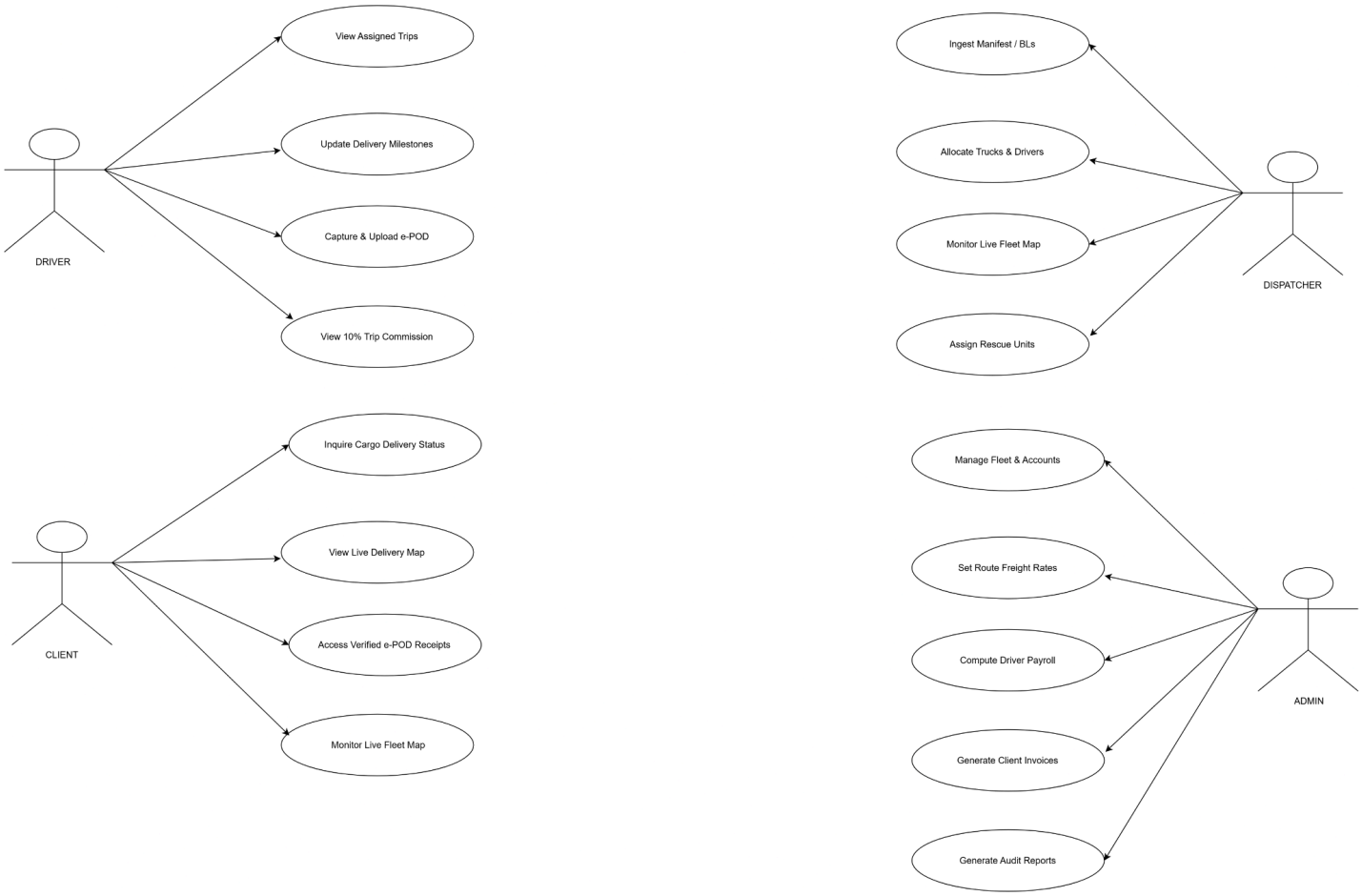


Figure B1. Use Case Diagram of the System

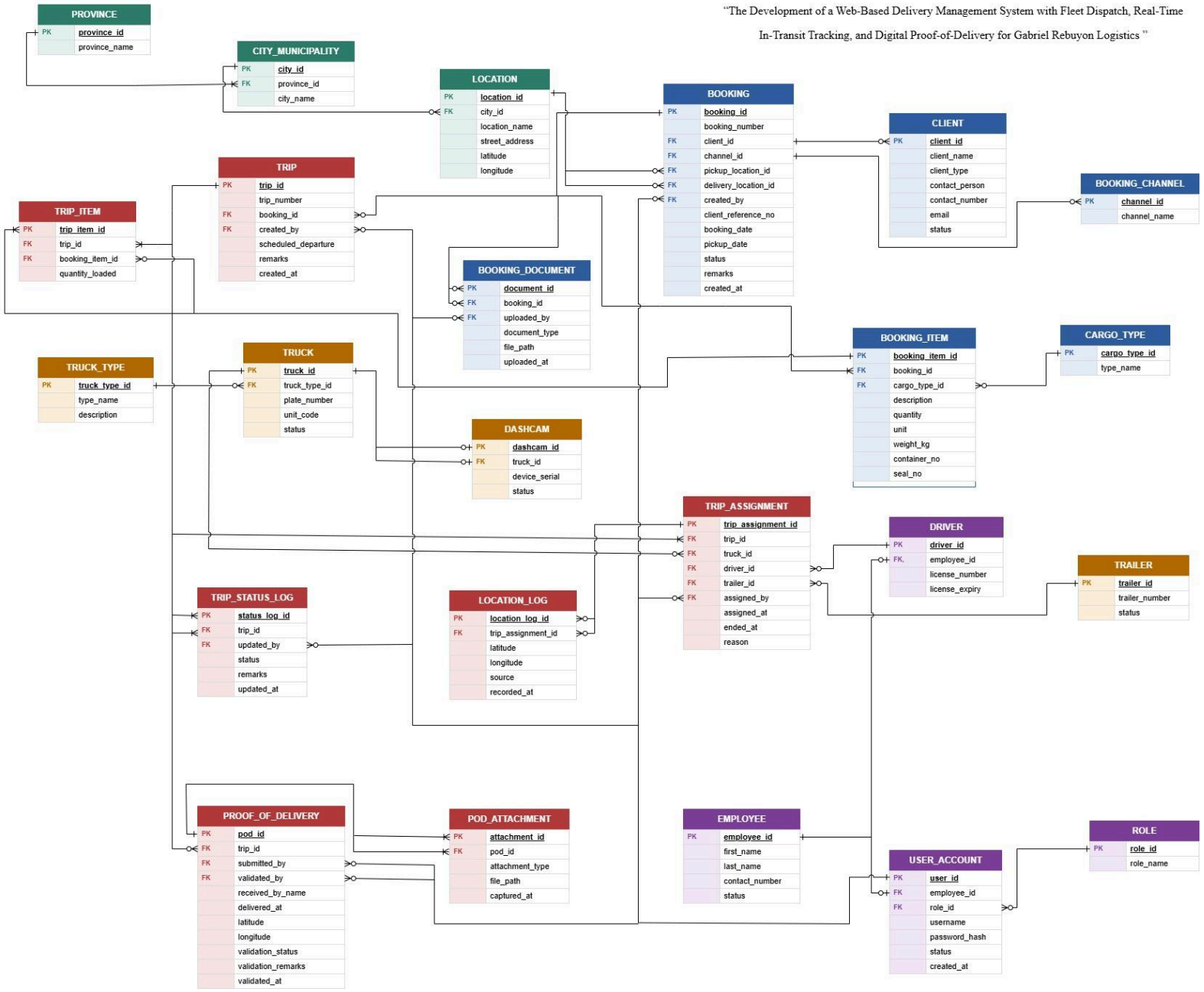
The Use Case Diagram depicts the functional interactions between the primary actors and the proposed fleet dispatch, tracking, and electronic proof-of-delivery platform. System operations are segmented across four key user roles: Driver, Client, Dispatcher, and Admin.

The **Driver** handles operational field tasks through a mobile-friendly view by viewing assigned trip schedules, transmitting progressive milestone updates throughout transit, capturing and uploading digital Electronic Proof of Delivery (e-POD) receipts, and checking accumulated earnings under the 10% trip commission structure. The **Client** accesses customer-facing visibility tools to submit cargo status inquiries, monitor active vehicle locations across transit routes on an interactive map, and view verified digital e-POD receipts.

The **Dispatcher** oversees daily freight operations by ingesting cargo manifests and incoming Bills of Lading (BL), assigning vehicles and available drivers, supervising all traveling fleet units via the live fleet telemetry map, and deploying standby rescue units when transit delays or vehicle breakdowns occur. Finally, the **Admin** manages core administrative and accounting modules, including maintaining vehicle and user account records, establishing route and container freight rate structures, calculating driver payroll commissions, issuing client billing statements, and generating system audit reports.

Appendix C – Entity-Relationship Diagram (ERD)

The ERD models the full delivery process of Gabriel Rebuyon Logistics. A CLIENT creates a BOOKING, which has pickup and delivery LOCATIONs (organized by PROVINCE and CITY_MUNICIPALITY) and one or more BOOKING_ITEMs classified by CARGO_TYPE. Each booking is dispatched as a TRIP, with TRIP_ITEMs recording the cargo loaded, and a TRIP_ASSIGNMENT linking the trip to a TRUCK, DRIVER, and TRAILER. Trip progress is monitored through TRIP_STATUS_LOG and LOCATION_LOG for real-time tracking, and completion is documented in PROOF_OF_DELIVERY with its POD_ATTACHMENTs. Access to the system is controlled through EMPLOYEE, USER_ACCOUNT, and ROLE.



Appendix D – System Architecture

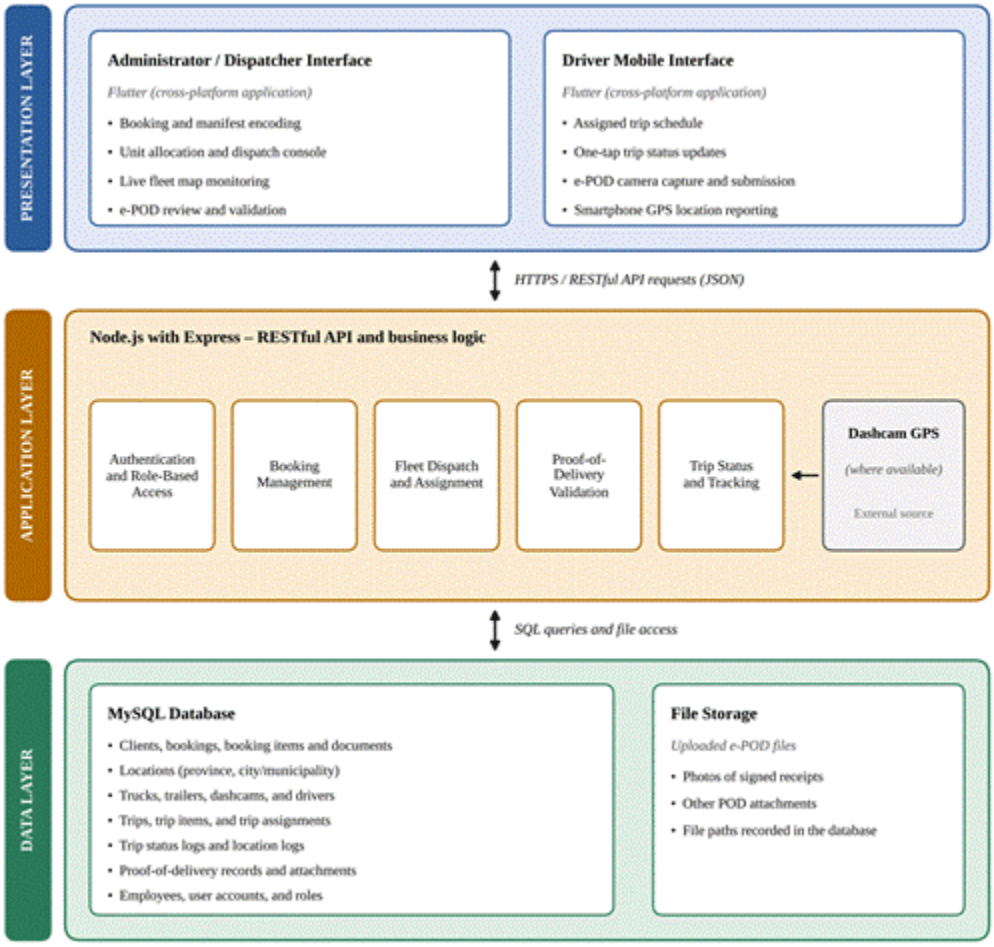


Figure D1. Three-Tier Architecture of the System

The system follows a three-tier architecture that separates the user interface, business logic, and data storage. In the presentation layer, a Flutter application provides the Administrator/Dispatcher interface (booking encoding, dispatch, fleet monitoring, and e-POD review) and the Driver mobile interface (assigned trips, status updates, e-POD submission, and GPS reporting). Both communicate with the application layer only through HTTPS RESTful API requests.

The application layer, built on Node.js with Express, authenticates users, enforces role-based access, and processes bookings, dispatch assignments, status and location updates, and proof-of-delivery validation. The Fleet Dispatch and Tracking Module, the core of the system, connects these functions so that every booking flows into dispatch and every delivery produces its own verifiable record. Dashcam GPS data is received here where available. Only

this layer accesses the data layer, where a MySQL database stores all system records and file storage keeps the e-POD attachments.

Appendix E – Figma UI/UX Prototypes

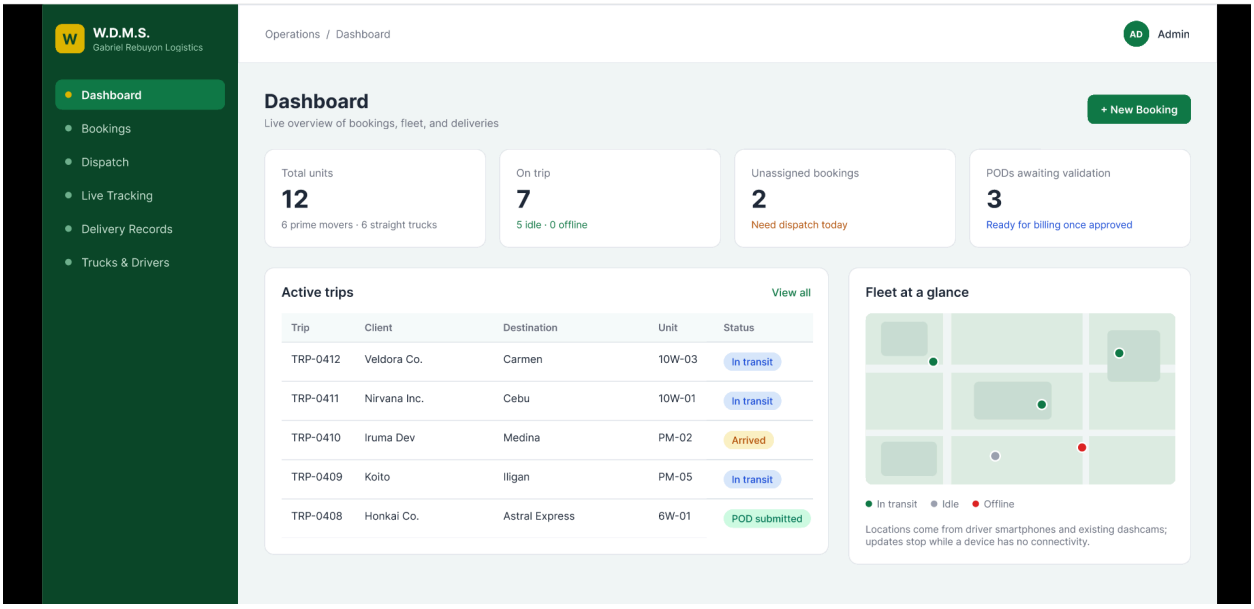


Figure E1. Admin Operations Master Dashboard Interface

This UI mockup demonstrates the planned design for the operational dashboard of Gabriel Rebuyon Logistics. It provides high-level executive metrics including total operating units (segmented into 6 prime movers and 6 straight trucks), units currently in transit versus idle, pending unassigned bookings, and e-PODs awaiting verification prior to billing consolidation.

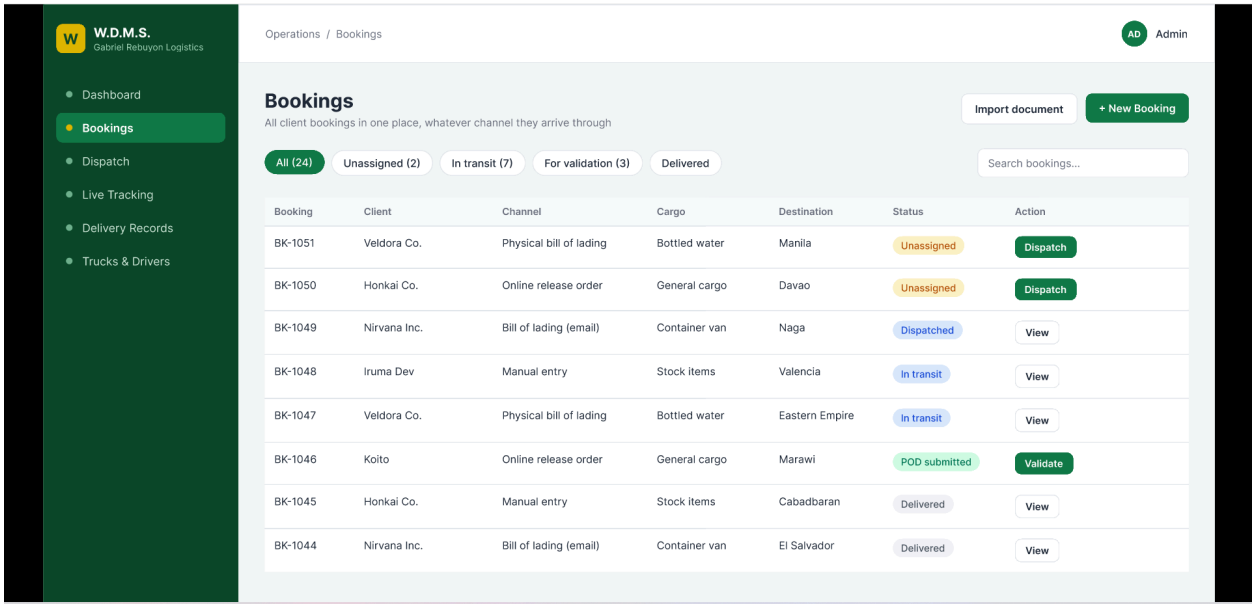


Figure E2. Consignment Booking and Manifest Ingestion Interface

This UI mockup demonstrates the planned design for the booking and consignment management console. It consolidates delivery orders arriving across multi-channel streams including physical bills of lading, forwarder email attachments, and online release orders enabling quick filtering by operational status.

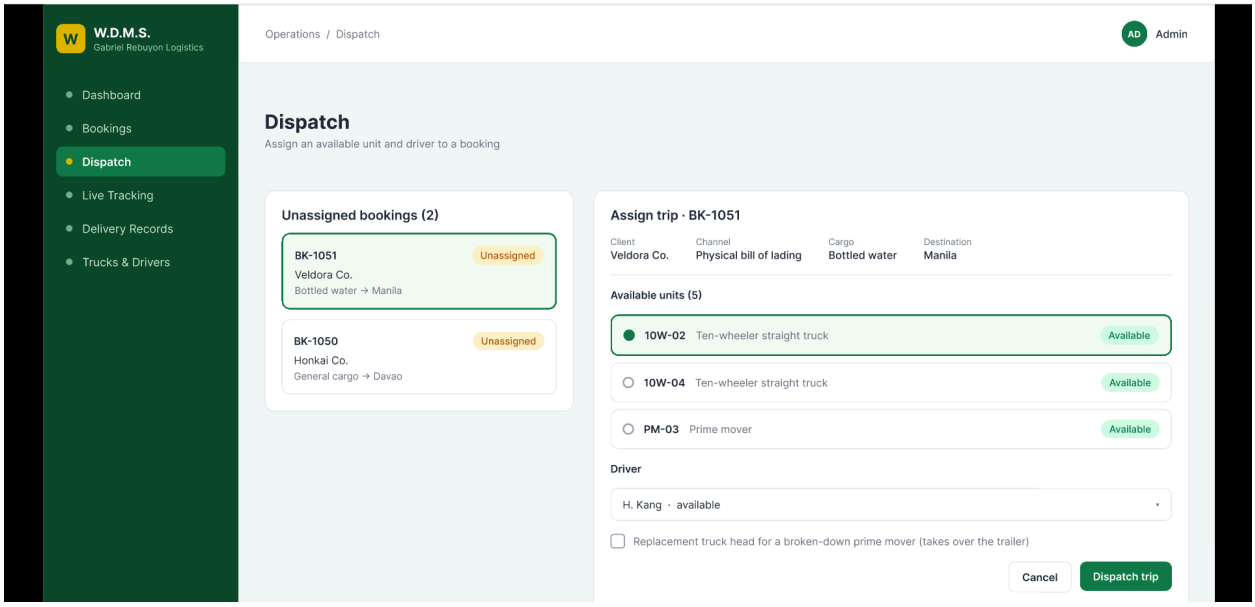


Figure E3. Unit Allocation and Rescue Dispatch Console

This UI mockup demonstrates the planned design for the dispatch allocation module. It pairs unassigned consignments with categorized vehicles (straight trucks vs. prime movers) and available standby drivers, featuring a dedicated contingency checkbox to dispatch replacement tractor heads for en-route breakdown rescue operations.

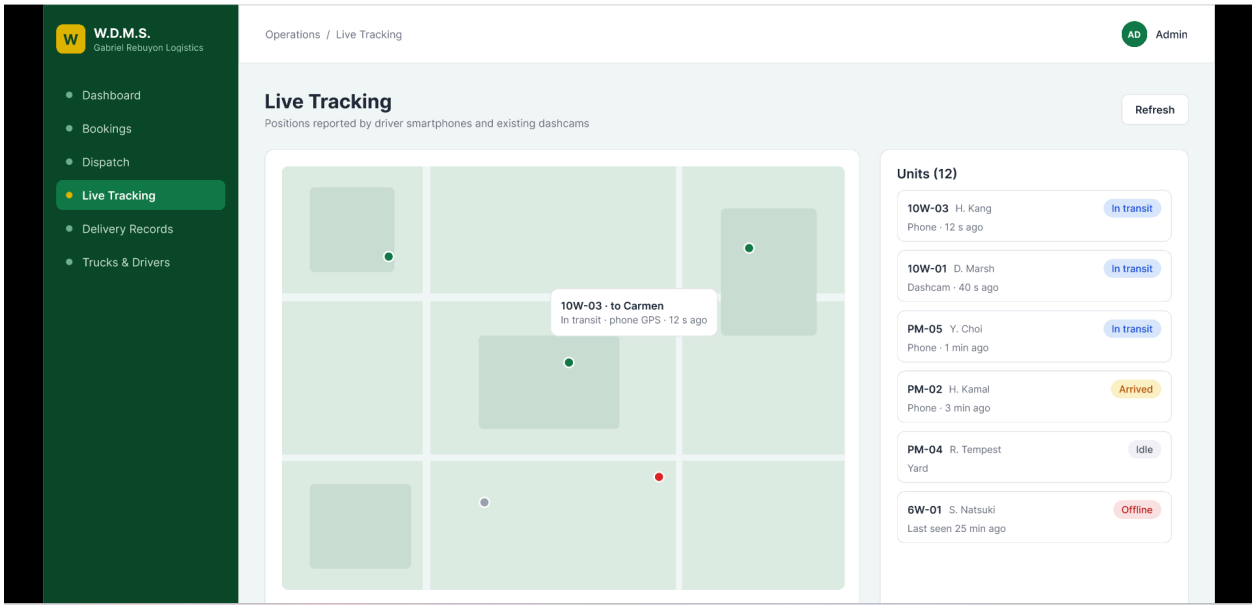


Figure E4. Live Telemetry Map and Fleet Monitor Interface

This UI mockup demonstrates the planned design for the dispatcher's live fleet monitoring interface. It plots real-time vehicle positions from mobile GPS pings and onboard dashcams, resolving client delivery status inquiries and flagging offline or idle assets.

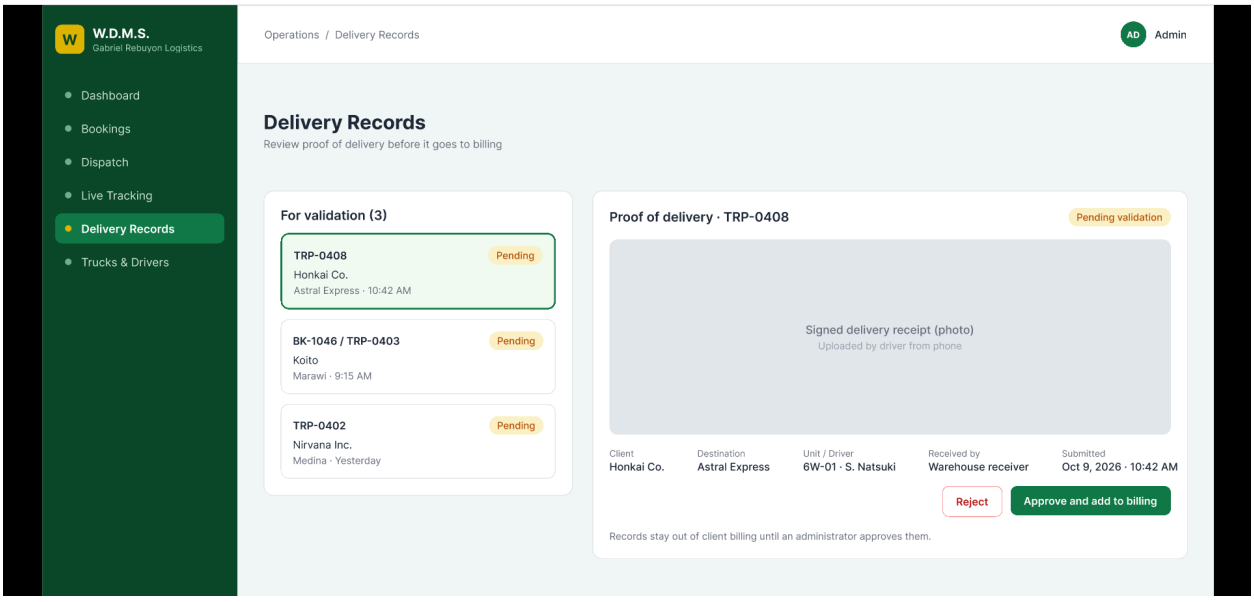


Figure E5. Delivery Records and Electronic Proof-of-Delivery Review Console

This UI mockup demonstrates the planned design for the administrative delivery records interface. Operations staff inspect and verify photographically submitted waybills from drivers prior to releasing the transaction to accounts receivable and driver commission payroll.

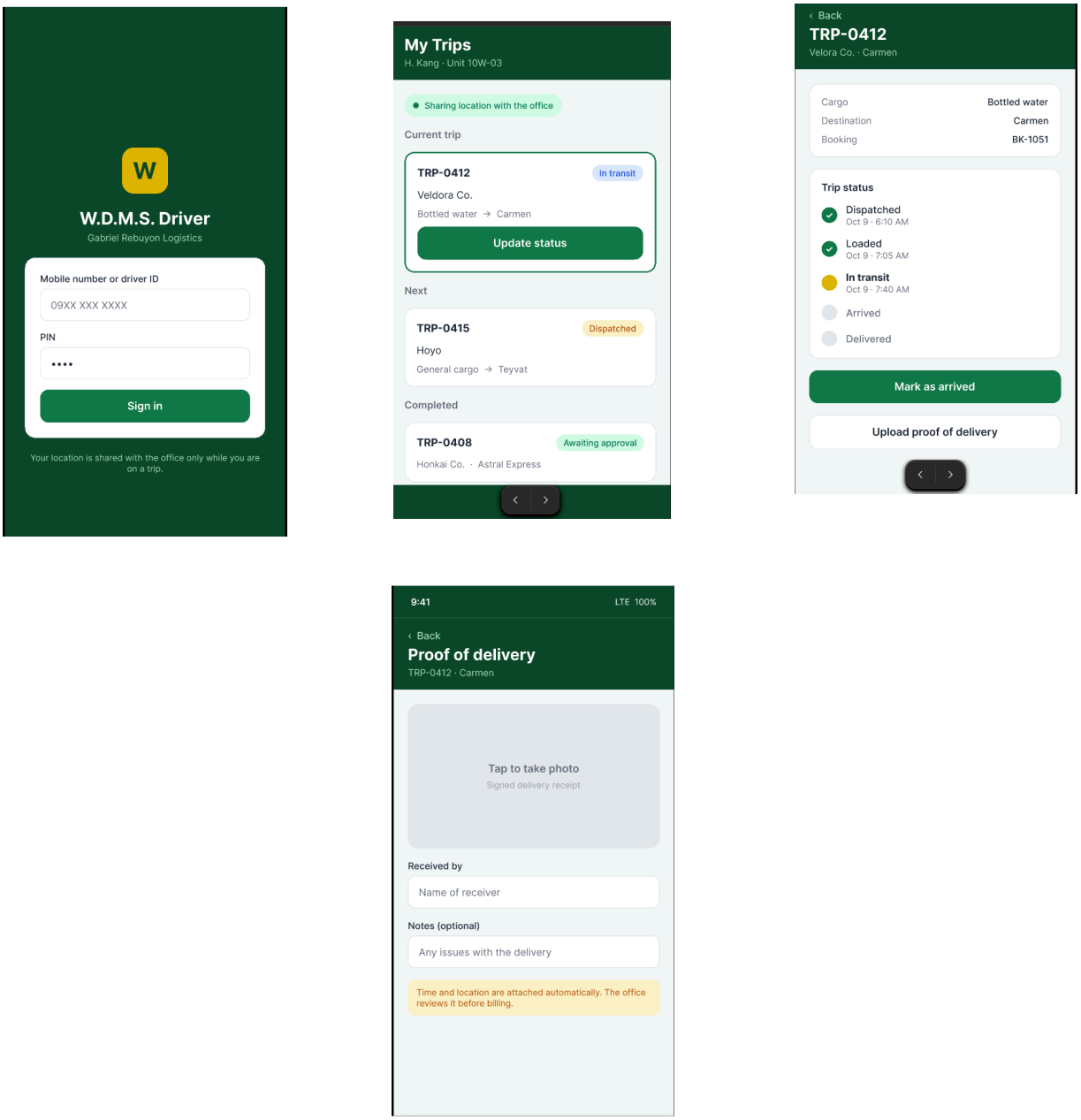


Figure E6. Driver Mobile Web Application Interfaces

This UI mockup demonstrates the planned mobile interface optimized for fleet drivers:

• (Screen A) Authentication: Secure, simplified driver login utilizing a registered mobile number and numeric PIN.

• (Screen B) Active Trip Manifest: Clear trip cards indicating active, queued, and completed runs alongside automated location-sharing indicators.

• (Screen C) Milestone Progression: One-tap status updates (*Dispatched*, *Loaded*, *In-Transit*, *Arrived*) that record verifiable timestamps for dispatch monitoring.

• (Screen D) e-POD Submission: Camera capture module allowing drivers to photograph signed physical delivery receipts, input recipient credentials, and send them directly to the billing department.

Appendix F – Testing Instruments

Appendix F.1 – Usability Evaluation Questionnaire

The usability questionnaire will be administered to the Administrator/Dispatcher and Drivers of Gabriel Rebuyon Logistics after they have used the developed system during pilot testing.

Respondent Information

Name (optional): _____ Date: _____

Role: ☐ Administrator/Dispatcher ☐ Driver

Instructions: Please rate each statement based on your experience using the system by placing a check (✓) in the box that matches your answer. Some statements are meant for specific respondents, as indicated in the headings. If a statement does not apply to your role, leave it blank.

Description	Rating
Strongly Agree	5
Agree	4
Neutral	3
Disagree	2
Strongly Disagree	1

A. Usability (Ease of Use)

No.	Evaluation Statement	5	4	3	2	1
1	The system is easy to learn and understand.	☐	☐	☐	☐	☐
2	The system's navigation is clear and organized.	☐	☐	☐	☐	☐
3	The screens, buttons, and labels are easy to read and use.	☐	☐	☐	☐	☐
4	Booking and trip information is easy to find and view.	☐	☐	☐	☐	☐
5	The information displayed by the system, such as trip status, tracking, and delivery records, is understandable.	☐	☐	☐	☐	☐

B. Functional Suitability

No.	Evaluation Statement	5	4	3	2	1
Booking and Dispatch (Administrator/Dispatcher)						
6	The system allows bookings to be recorded completely, including the client, pickup and delivery locations, and cargo details.	☐	☐	☐	☐	☐
7	The system allows trips to be created from bookings and cargo to be assigned to each trip.	☐	☐	☐	☐	☐
8	The system allows a truck, driver, and trailer to be assigned to a trip correctly.	☐	☐	☐	☐	☐
Trip Tracking (all respondents)						
9	The system displays the current status of a trip clearly.	☐	☐	☐	☐	☐
10	The real-time tracking shows the truck's location correctly while in transit.	☐	☐	☐	☐	☐
11	Trip status updates are recorded and can be viewed afterward.	☐	☐	☐	☐	☐
Proof of Delivery (all respondents)						
12	The system allows proof of delivery, including photos and delivery details, to be submitted and viewed properly.	☐	☐	☐	☐	☐
13	The proof of delivery shows the receiver, delivery time, and location correctly.	☐	☐	☐	☐	☐
Proof of Delivery Validation (Administrator/Dispatcher)						
14	The system allows submitted proofs of delivery to be reviewed and validated.	☐	☐	☐	☐	☐

C. Performance Efficiency

No.	Evaluation Statement	5	4	3	2	1
15	The system loads screens and information quickly.	☐	☐	☐	☐	☐
16	The tracking information updates within a reasonable amount of time.	☐	☐	☐	☐	☐
17	Recording bookings, dispatching trips, and submitting proof of delivery take little time.	☐	☐	☐	☐	☐
18	The system saves time compared with the current manual process of booking, dispatching, and delivery documentation.	☐	☐	☐	☐	☐

D. Reliability and Accuracy

No.	Evaluation Statement	5	4	3	2	1
19	The system runs without errors or unexpected crashes.	☐	☐	☐	☐	☐
20	The truck, driver, and trailer assignments contain no conflicting or duplicated entries.	☐	☐	☐	☐	☐
21	The GPS location recorded by the system is accurate.	☐	☐	☐	☐	☐
22	Saved bookings, trip records, and proof of delivery stay accurate and complete when viewed again.	☐	☐	☐	☐	☐
23	The system accurately records trip status updates and delivery times.	☐	☐	☐	☐	☐

E. Security and Access Control

No.	Evaluation Statement	5	4	3	2	1
24	The system requires users to log in with their own account.	☐	☐	☐	☐	☐
25	The system gives each user only the features appropriate to their role (Administrator/Dispatcher or Driver).	☐	☐	☐	☐	☐
26	Booking, trip, and client information is protected from unauthorized users.	☐	☐	☐	☐	☐
27	The system shows appropriate messages when incorrect information is entered.	☐	☐	☐	☐	☐

F. Portability

No.	Evaluation Statement	5	4	3	2	1
28	The system opens and runs properly on my device (computer or mobile phone).	☐	☐	☐	☐	☐
29	The system displays properly on my device's screen.	☐	☐	☐	☐	☐
30	The system works properly in the places where I use it (office or delivery routes).	☐	☐	☐	☐	☐

G. Overall Satisfaction

No.	Evaluation Statement	5	4	3	2	1
31	I am satisfied with the overall functionality of the system.	☐	☐	☐	☐	☐
32	The system meets the delivery management needs of Gabriel Rebuyon Logistics.	☐	☐	☐	☐	☐
33	I would recommend the system for actual use by Gabriel Rebuyon Logistics.	☐	☐	☐	☐	☐
34	The system is an improvement over the current manual process of booking, dispatching, tracking, and proof of delivery.	☐	☐	☐	☐	☐

Comments and Suggestions